

PRIMEASIA UNIVERSITY

Department of Computer Science & Engineering

FINAL PROJECT REPORT

ON

SMART ATTENDANCE SYSTEM

Submitted By	Student ID
Md. Showeb Akhter Srabonto	242-005-042
Tania Akter	242-025-042
Anon Dutt	242-001-042

Course Code	Course Title
CSE-232	Database System Design (Lab)

Submitted To: Miss Halima Mowla, Lecturer — Department of Computer Science & Engineering,
Primeasia University

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Abstract

The Smart Attendance System is a web-based application developed using PHP and MySQL to automate attendance and student result management in educational institutions. The system features role-based access with separate secure logins for administrators and students. Administrators can perform CRUD operations on student records, manage daily attendance, record course marks, and generate reports on attendance percentages and academic performance. Students can log in to view their personal attendance history, marks, and profile information. The system eliminates manual paperwork, reduces human errors, and provides a centralized digital database for reliable data storage and retrieval. While currently deployed on a local XAMPP server with manual attendance entry, the system successfully demonstrates practical database design concepts. Future improvements include QR code integration, facial recognition, and cloud deployment for enhanced scalability.

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1. Introduction

Background of the Project

Smart Attendance System is a web-based management system developed using PHP, MySQL, HTML, and CSS. The main purpose of this project is to simplify attendance management and student result management in educational institutions. The system includes separate login systems for Admin and Students. Admin can manage attendance, marks, reports, and students. Students can log in and view their personal marks and profile information.

Problem Statement

- ❖ Attendance sheets can be lost or damaged.
- ❖ Calculating attendance percentage manually is difficult.
- ❖ Managing student marks manually takes time.
- ❖ Students cannot easily view their marks.
- ❖ Teachers face difficulties maintaining records.
- ❖ Human errors occur frequently.

Objectives

- ❖ To automate the attendance management process.
- ❖ To reduce manual paperwork.
- ❖ To provide a secure login system.
- ❖ To store attendance and marks digitally.
- ❖ To create a student portal.
- ❖ To generate reports easily.
- ❖ To improve data management using databases.
- ❖ To provide a modern and user-friendly interface.

2. Requirement Analysis

Functional Requirements

- ❖ Admin login and session management.
- ❖ Add, update, and delete student records.
- ❖ Take and manage attendance (present/absent).
- ❖ Add, update, and delete marks records.
- ❖ Generate attendance, marks, and percentage reports.
- ❖ Student registration and secure login.
- ❖ Student portal to view personal marks and profile.

Non-Functional Requirements

Requirement	Description
Performance	Fast page loads on local server (XAMPP)
Security	Role-based access with PHP session authentication
Usability	Simple, clean, and intuitive UI for all user types
Reliability	Consistent data stored in MySQL database
Scalability	Modular design allows easy feature addition
Compatibility	Works on all modern browsers

3. System Design

System Architecture

The Smart Attendance System follows a three-tier architecture: the Presentation Layer (HTML/CSS frontend), the Application Layer (PHP backend logic), and the Data Layer (MySQL database). The XAMPP server acts as the local hosting environment that integrates Apache and MySQL.

Workflow

Step	Description
User Login	Admin or Student logs in via the authentication module.
Role Check	PHP session verifies role and redirects to appropriate dashboard.
Admin Actions	Admin manages students, attendance, marks, and reports.
Student Actions	Student views profile and personal marks.
Report Generation	System queries MySQL and displays attendance/marks summaries.
Logout	Session is destroyed and user is redirected to login.

Module Description

- ❖ Authentication
- ❖ Attendance
- ❖ Student Management
- ❖ Marks Management
- ❖ Report Module
- ❖ Student Portal

4. Database Design

ER Diagram Description

Database Name: attendance_db

- ❖ One student can have many attendance records.
- ❖ One student can have many marks records.
- ❖ Student ID is used as a Foreign Key in both Attendance and Marks tables.
- ❖ Users table stores login credentials (username, password, role).

Database Schema

Table	Fields	Description
users	id (PK), username, password, role	Stores admin and student login credentials
students	student_id (PK), name, department	Stores student profile information
attendance	id (PK), student_id (FK), status, date	Stores daily attendance records
marks	id (PK), student_id (FK), course_name, mid_marks, final_marks	Stores exam marks per course

Use Cases

Admin: Login, Add Student, Take/Delete Attendance, Add/Delete Marks, View Reports.

Student: Register, Login, View Profile, View Personal Marks.

5. Implementation

Category	Technologies / Tools
Frontend Technologies	HTML5, CSS3, JavaScript
Backend Technologies	PHP (Procedural)
Database	MySQL via phpMyAdmin
Frameworks / Libraries	None (Pure PHP & CSS)
Development Environment	XAMPP, VS Code
Server	Apache (local via XAMPP)
Hardware Requirements	Intel Core i3+, 4GB RAM+, Windows/Linux

Key Code Snippets:

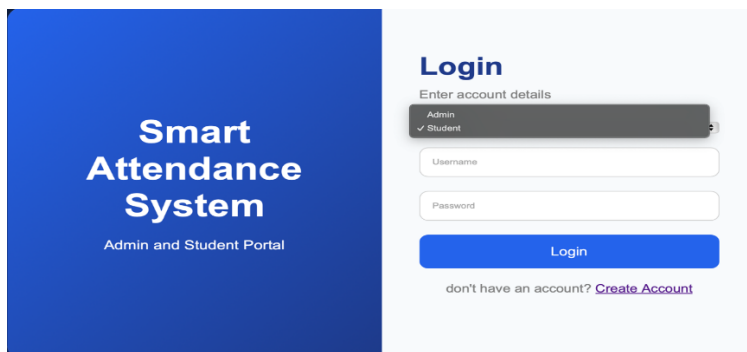
Feature	Code
Database Connection	<code>mysqli_connect("localhost", "root", "", "attendance_db")</code>
Session System	<code>session_start();</code>
Fetch Data	<code>mysqli_fetch_assoc(\$result)</code>
SQL Query Execution	<code>mysqli_query(\$conn, \$sql)</code>

6. User Interface

The following screens form the user interface of the Smart Attendance System.

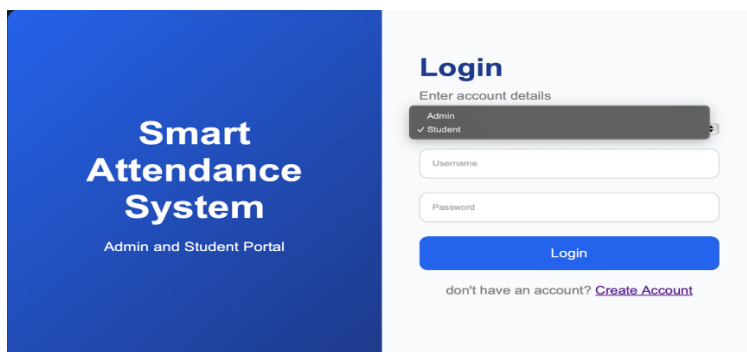
• Home Page

Landing page with navigation links to Admin Login and Student Login.



• Login Page

Secure login form with username/password fields. Role determined automatically.



• Admin Dashboard

Summary panel with quick links to Attendance, Students, Marks, and Reports.

The screenshot shows the Admin Dashboard with a sidebar menu on the left and a main content area. The sidebar menu is titled 'Attendance' and includes links for Dashboard, Report, Percentage, + Add Student, Add Marks, Marks Report, and Logout. The main content area is titled 'Add Student' and contains a form with three input fields: Student ID, Student Name, and Department. Below the input fields is a blue button labeled 'Add Student'.

• CRUD Operations

Add/Delete Students, Take/Delete Attendance, Add/Delete Marks via forms.

The screenshot shows the Admin Dashboard with a sidebar menu on the left and a main content area. The sidebar menu is titled 'Attendance' and includes links for Dashboard, Report, Percentage, + Add Student, Add Marks, Marks Report, and Logout. The main content area is titled 'Add Marks' and contains a form with four input fields: Student ID, Course Name, Mid Exam Marks, and Final Exam Marks. Below the input fields is a blue button labeled 'Add Marks'.

• Reports

Attendance Report, Marks Report, and Percentage Report available.

05/13/2026 Search All

Name	Date	Status	Action
Lamia	2026-05-11	Present	Delete
Anon Dutt	2026-05-10	Present	Delete
Tania Akter	2026-05-10	Present	Delete
Lamia	2026-05-09	Present	Delete
Arif	2026-05-09	Present	Delete
Sayma Hossen	2026-05-07	Present	Delete
Hasan	2026-05-06	Present	Delete
Hossain	2026-05-06	Present	Delete
Imran	2026-05-05	Absent	Delete
Lamia	2026-05-05	Absent	Delete
Tanvir	2026-05-05	Present	Delete
Mizan	2026-05-05	Present	Delete

Attendance

- Dashboard
- Report
- Percentage
- + Add Student
- Add Marks
- Marks Report
- Logout

Marks Report

ID	Student ID	Course	Mid	Final	Total	Grade	Action
1	242047042	Database Management	11	40	51	C	Delete
2	242047042	Database Management	11	40	51	C	Delete
4	242001042	Database	7	39	46	F	Delete
6	242025042	numerical	22	50	72	A	Delete
7	242047042	Numerical	22	30	52	C	Delete
8	242006042	Numercial	20	30	50	C	Delete
9	242025042	TOC	47	40	87	A+	Delete
10	242008042	Numerical	35	40	75	A	Delete
11	242053042	EEE	35	39	74	A	Delete

• Student Portal

Students view personal attendance and marks in a read-only secure view.

Smart Attendance System

Admin and Student Portal

Login

Enter account details

Admin
☒ Student

Username

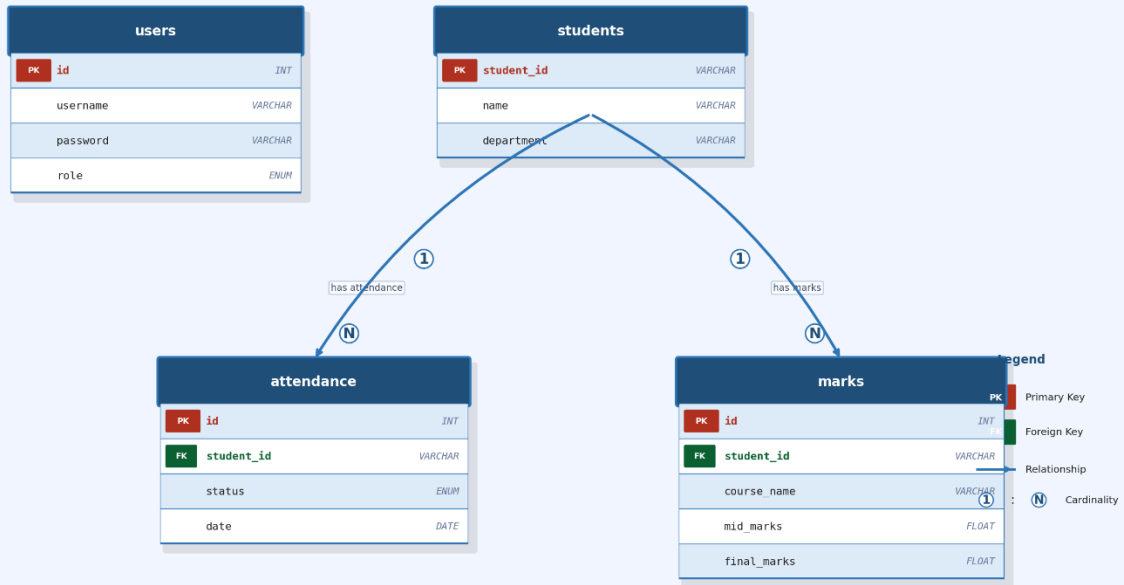
Password

Login

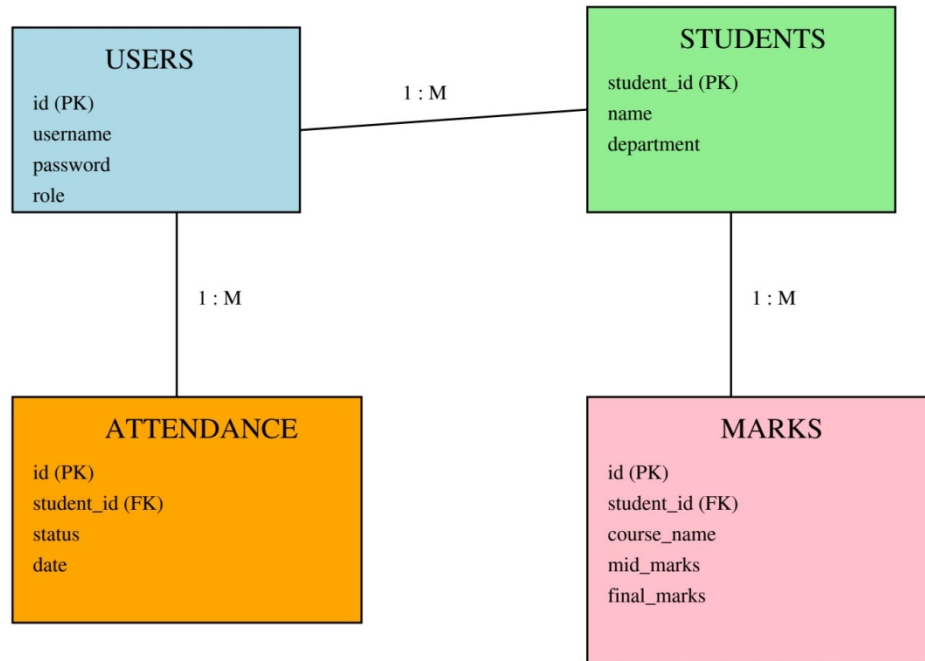
don't have an account? [Create Account](#)

Smart Attendance System — Database Schema

Database: attendance_db | CSE-232 Final Project



Smart Attendance System ER Diagram



Explanation:

USERS table stores login information.
STUDENTS table stores student details.
ATTENDANCE table stores attendance records.
MARKS table stores exam marks.
student_id is used as a Foreign Key.

10. Challenges & Limitations

- ❖ Works on a local server only (XAMPP); not deployed on a live web server.
- ❖ No email or SMS notification system for students.
- ❖ No mobile application — accessible only via desktop browsers.
- ❖ No biometric or QR-code-based attendance; attendance is entered manually.
- ❖ Password storage does not currently use hashing (security improvement needed).
- ❖ No real-time data synchronization across multiple devices.

11. Future Improvements

- ❖ QR Code-based automated attendance tracking.
- ❖ Face Recognition attendance system using AI/ML.
- ❖ Dedicated Mobile Application (Android/iOS).
- ❖ Migration to a Cloud Database for remote access.
- ❖ Email Notifications for attendance alerts.
- ❖ SMS Alerts to parents/students for low attendance.
- ❖ Password hashing and enhanced security features.
- ❖ Export reports to PDF/Excel directly from the system.

12. Conclusion

The Smart Attendance System successfully automates attendance and marks management using PHP and MySQL. The project significantly reduces manual work, minimizes human error, and provides a secure and user-friendly platform for both administrators and students. The system fulfills all functional requirements outlined in the project format and demonstrates a practical application of database system design concepts learned in CSE-232. With future enhancements such as QR-code attendance, mobile support, and cloud deployment, this system has the potential to serve as a fully scalable institutional management solution.

References

- ❖ PHP Official Documentation – <https://www.php.net/docs.php>
- ❖ MySQL Official Documentation – <https://dev.mysql.com/doc/>
- ❖ W3Schools – <https://www.w3schools.com>
- ❖ MDN Web Docs – <https://developer.mozilla.org>
- ❖ XAMPP Documentation – <https://www.apachefriends.org>